

Suraj Kumar

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EDUCATION

Case Western Reserve University (CWRU)

Cleveland, OH

Combined B.S./M.S. Program

Expected Graduation: Dec 2027

Master of Science (M.S.), Computer Science

Aug 2026 – Dec 2027

Bachelor of Science in Engineering (B.S.E.), Biomedical Engineering

Aug 2023 – Dec 2027

Relevant Coursework: Operating Systems (in C), Intro to AI, Algorithms, Data Structures (in Java), Discrete Math

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, HTML/CSS, SQL, Java, C

Frameworks & Libraries: Django, FastAPI, SQLAlchemy, React.js, React Native, Next.js, PyTorch, TensorFlow, Scikit-learn, vLLM, HuggingFace, OpenCV, Selenium

Tools: Git, AWS (ECS, VPC, RDS, S3, KMS), Docker, CI/CD, PostgreSQL, Linux, REST APIs, WebSockets

WORK/RESEARCH EXPERIENCE

AI Security Researcher

May 2026 – Present

Case Western Reserve University PEAT AI Lab

Remote/Cleveland, OH

- Built an unsupervised anomaly detection framework that flags intrusions in streaming network-flow logs at **>0.98 AUROC** and **<3% FPR**, scoring each window against a learned geometric representation of normal traffic.
- Identified the top-performing intrusion-detection configuration out of 16 candidates by benchmarking embedders (TF-IDF, Word2Vec, Qwen3) against detectors (GMM, KDE, OCSVM, DeepSVDD) across **727 experiments**.
- Scaled embedding generation across distributed Linux HPC nodes using chunked streaming I/O and memory-efficient attention, **processing 10M+ flow records** while **cutting peak GPU memory 10x**.

Machine Learning Engineer Intern

Jun 2025 – Dec 2025

Genentech

South San Francisco, CA

- Reduced the Bioassay department's experimental workload by 30%** by training machine learning models (PyTorch and Scikit-learn) to predict which drug-potency experiments would fail and flag redundant ones.
- Deployed ML inference endpoints to AWS EKS via GitLab CI/CD pipelines with automated pytest suites, serving **~200 predictions/day** to analytical scientists running lab experiments.
- Cut training-data retrieval time from **2+ hours to 5 minutes** by consolidating 30+ unorganized SQL tables.
- Drove adoption of the analytics tool across **3 departments and 110 employees** through technical demos.

Deep Learning Researcher

Sep 2024 – Nov 2025

Case Western Reserve University INVent Lab

Remote/Cleveland, OH

- Engineered a deep learning pipeline for glomeruli segmentation in pathology images that leverages a foundation model (MedSAM) to provide pseudo-labels for a UNet, **increasing baseline accuracy by 19%**.
- Trained an EfficientNet-B3 binary classifier to filter non-glomeruli tiles at **94% precision**, reducing downstream segmentation noise and **shrinking the testing pool by 40%** before MedSAM labeling.
- Refined ground truth masks via morphological opening and connected-component area filtering, decreasing spurious segmentations by **45%** to raise pseudo-label quality across training iterations.

SOFTWARE ENGINEERING PROJECTS

[AcustiCare](#) | Software Engineering Lead | AWS, Docker, Django, React Native, Expo EAS Nov 2025 – Present

- Led development of a voice-screening mobile app for ENT physicians that automates audio analysis of patient vocal health, cutting per-patient screening from a **2-hour clinic visit** to an at-home recording processed in **seconds**.
- Architected HIPAA-compliant AWS infrastructure with a signed BAA, network-isolated RDS, load-balancing, and pre-signed S3 URLs for audio access, protecting clinical data and processing **120 recordings/week** at peak.
- Translated Figma designs into a React Native frontend through iterative physician feedback, engineering TanStack Query caching across **15 app screens** to cut repeat-visit render latency by **90%** (~ 250 ms $\rightarrow$ ~ 25 ms).

[Rival](#) | Software Engineering Lead | React Native, FastAPI, Gemini API, OAuth 2.0 Jan 2026 - May 2026

- Built a 1v1 productivity mobile game where a Gemini AI referee scores GitHub and Notion activity to decide each match's winner, integrating **2 OAuth providers** and **31 REST endpoints**.
- Cut up to 8s of latency** by replacing a polling loop with a WebSocket push layer that streams AI-referee verdicts and live opponent progress to both players the instant that match state changes.